# Build or Subsidize?

## The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband

Sidi Sawadogo

Université de Montréal and CIREQ

Séminaire Marcel-Dagenais
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# Motivation: Two barriers to broadband access

Constraint to broadband access

- **Affordability:** service exists, but some households cannot afford it
  15% of broadband users report trouble paying; 34% below $30,000 income (Pew, 2021)
- **Deployment:** high fixed costs leave some locations unserved
  21.3 million Americans lacked 25/3 Mbps fixed broadband at end-2017 (FCC, 2019)

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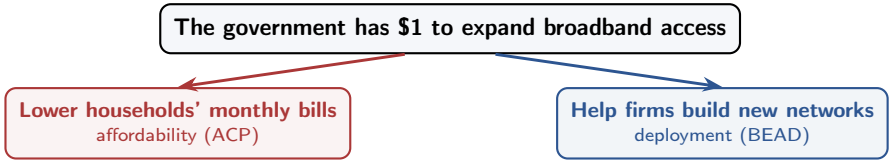
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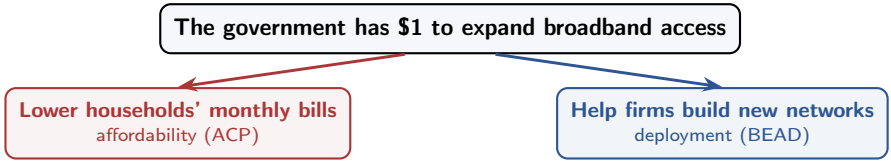
**Few local providers:** 63.7% of tracts have at most two providers, so the effects of subsidies may depend on how firms respond

▸ A1: Program timeline

# This paper: research question

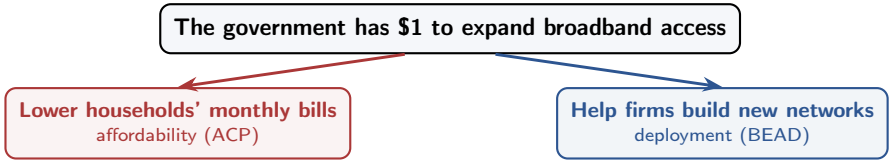


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**Approach**

1. **Event studies:** Examine how markets responded to ACP and BEAD
2. **Structural model:** Estimate household demand and model firms' pricing and product choices
3. **Counterfactuals:** Compare welfare effects per public dollar

# Preview of findings

1. **Reduced form: market responses**
   - ACP: Lowered posted prices by 7–10% and raised the high-speed share by 6–11pp (P25 vs. P75 exposure)
   - BEAD: Limited evidence of early responses in the observed sample
2. **Structural estimates**
   - Elastic product-level demand: $-4.99$; median margin of 20%
   - Fixed costs are large and only partially identified span approximately $\$60k - \$1M$ per year
3. **Welfare per public dollar**
   - ACP: about \$1 per dollar each year, largely a transfer
   - BEAD: about \$7 per grant dollar over five years (\$1.5 per year)

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**Policy implication:** the right instrument depends on the local barrier to access

# Contributions and related literature

1. **A welfare comparison of affordability and deployment subsidies**

   Compare the two instruments nationally, per public dollar, allowing entry and product choices to respond

   Closest work: Kearns (2026, working paper); Fan and Xiao (2015); Ackerberg et al. (2014)

2. **Fixed-cost bounds with many providers and product choices**

   I adapt profit-bound moment inequalities to 1,732 providers and 70,854 tracts

   Closest work: Ciliberto and Tamer (2009, ECTA); Eizenberg (2014, REStud); Ciliberto et al. (2021, JPE); Fan and Yang (2025, JPE)

3. **Evidence for matching subsidies to local access barriers**

   Deployment gains are largest in monopoly tracts; affordability gains are largest in low-income tracts

   Closest work: Espín and Rojas (2024, IJIO); Kearns (2026, working paper)

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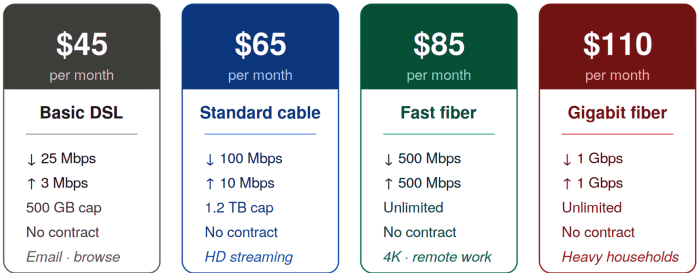
5) Counterfactuals

6) Conclusion

# Industry Background and Data

# Broadband: differentiated products and local competition

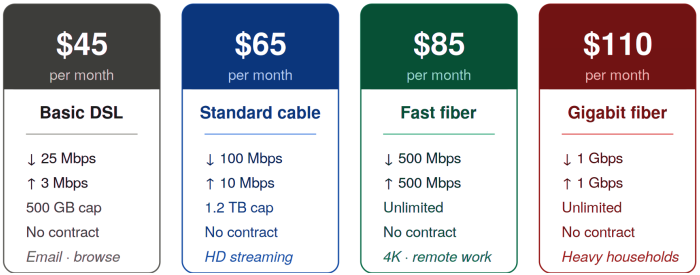
**What do households choose?** Plans differ in speeds, data allowances, and prices




Illustrative plans

# Broadband: differentiated products and local competition

**What do households choose?** Plans differ in speeds, data allowances, and prices




Illustrative plans

**Who provides broadband?**

- Telecom and cable operators; some municipal networks
- 1,732 providers; one or two in most tracts

**Technologies**

| | | | |
|---|---|---|---|
| **DSL** | Copper phone lines | **Fiber** | Optical fiber |
| **Cable** | Coaxial cable | **Fixed wireless** | Radio links |

# Data: a new tract-level dataset of U.S. broadband

**Sources**

- **Federal Communications Commission (FCC) Urban Rate Survey:** plan prices, speeds, technology (urban areas)
- **FCC Form 477:** which providers offer which speed tiers
- **FCC adoption data:** tract-level broadband adoption
- **American Community Survey (ACS), Census Bureau:** income, poverty, households
- **National Telecommunications and Information Administration (NTIA):** BEAD allocations by state

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**Two samples**

- **Event studies:** state-by-year panel, 52 jurisdictions, 2014–2025 (624 state-years)
- **Structural model:** 727,601 potential provider–speed-tier positions in 70,854 tracts

▸ Descriptive statistics ▸ B1: Sources ▸ B2: Shares ▸ B4: Samples

# Descriptive statistics: the typical tract and market structure

**The typical tract**

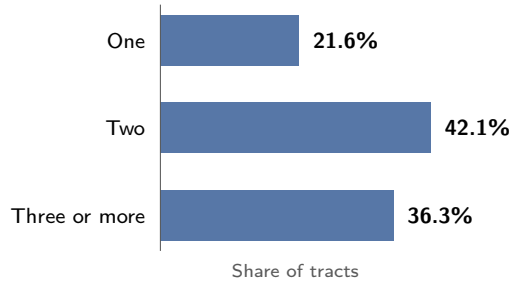
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| Average monthly price | **$74** |
| Average population | **4,533** |
| Median household income | **$65,422** |
| Households with no broadband | **19.1%** |

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**Providers per tract**




- Median within-tract HHI: **5,472**

Data | B1: Sources | B2: Shares | B3: Coverage

# Reduced-Form Evidence

# Empirical design: Continuous-treatment event study

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\tau \neq -1} \beta_\tau \left( D_i^P \times 1\{t - \bar{t} = \tau\} \right) + \gamma_i t + \varepsilon_{it}, \quad P \in \{\text{ACP}, \text{BEAD}\} \tag{1}$$

- ACP exposure $D_i^{ACP}$: 2020 share of households below 200% of the federal poverty line
- BEAD exposure $D_i^{BEAD}$: log formula allocation per unserved location
- $Y_{it}$: average price, product mix, or provider HHI in state $i$
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- State and year fixed effects, state trends; weighted by plans; SEs clustered by state

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**Identification**: Absent the programs, outcomes in more- and less-exposed states would have followed similar trends, conditional on fixed effects and state-specific trends

▸ C1: Exposure maps ▸ C2: Detail ▸ C3: Anticipation

# ACP lowers prices and shifts offerings toward faster plans



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**P25 → P75 exposure:** price −7 to −10%; high-speed price −15 to −18%; high-speed share +6 to +11pp

Coefficients per unit of exposure (share below 200% FPL; P25 → P75 = 0.10); $\tau = -1$ omitted; 95% CIs, s.e. clustered by state.

# BEAD: Limited evidence of early responses

- **Pre-trends**: no clear differential trends
- **Prices**: no detectable change, for low- or high-speed products
- **Product mix**: gradual shift toward high-speed products, not significant
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- **Takeaway**: imited evidence in the early period. The structural model evaluates deployment subsidies

▸ C6: BEAD event study ▸ C5: Pooled DiD ▸ C7: Urban frame

# Why do I need a structural model?

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- **Welfare:** who gains, by how much, and per public dollar
- **BEAD once networks are built:** entry where no provider operates today
- **Mechanisms:** new adoption vs. business stealing vs. entry
- **Other designs:** subsidy levels not observed ($10–$30 discounts; 25–75% of fixed costs)

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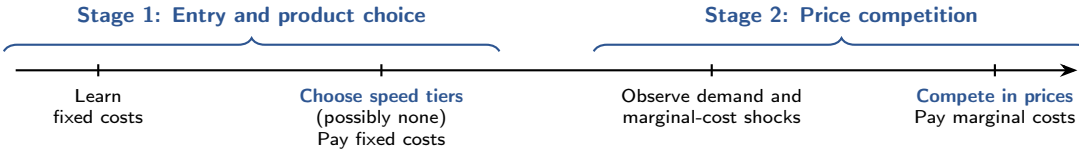
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$\Rightarrow$ **A model of household demand, provider pricing, and entry**, estimated on 70,854 tracts

# Structural Model

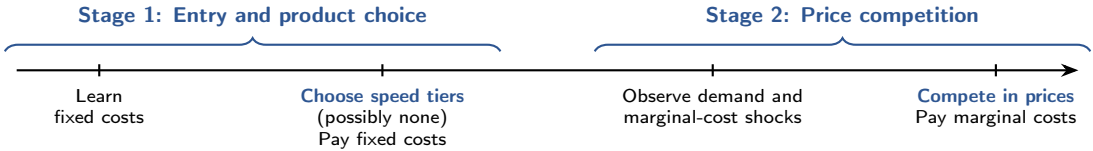
# Structural model in a nutshell

Providers play a two-stage game in each census tract.



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**Estimation: work backward**

1. Estimate household demand and recover marginal costs
2. Compute expected variable profits across product portfolios
3. **Multiple entry equilibria** $\Rightarrow$ bound fixed costs (partial identification)

Fan and Yang (2025, JPE); Andrews and Soares (2010, ECTA)

Point-identified: demand, markups. Bounded: fixed costs. Assumed: nested logit, Bertrand-Nash pricing, timing.

▸ Skip to counterfactuals

# Second stage: Demand and pricing

# Model of household demand for broadband

- **Market:** census tract $i$
- **Product $j$:** a provider's speed tier (e.g., Verizon, high-speed tier)
- **Nest:** all broadband products in one nest; no broadband is the outside option (nested logit)

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Household $h$ in tract $i$ chooses a product $j$, or the outside option ($U_{h0i} = \varepsilon_{h0i}$)

$$U_{hji} = \underbrace{X_{ji}^T \beta + \xi_{ji} + \alpha\, p_{ji}}_{\delta_{ji}\text{: mean utility}} + \nu_{h,\text{in}} + (1-\rho)\,\varepsilon_{hji}$$

- $X_{ji}$: download and upload speeds; $\xi_{ji}$: unobserved quality; $p_{ji}$: monthly price ($100)
- $\nu_{h,\text{in}}$: household taste for broadband, common to all products in the nest
- $\varepsilon_{hji}$: household taste for product $j$, i.i.d. type I extreme value; $\rho \in [0, 1)$: how alike products are

▸ D1: Why one nest

# Estimation and identification of demand

**Estimation:** invert observed shares into a linear equation (Berry, 1994 RAND)

$$\underbrace{\ln s_{ji} - \ln s_{0i}}_{\text{data}} = X_{ji}^T \beta + \alpha\, p_{ji} + \rho \ln s_{j|\text{in},i} + \underbrace{\xi_{ji}}_{\text{error}}$$

$s_{ji}$, $s_{0i}$: shares of product $j$ and of no broadband; $s_{j|\text{in},i} = s_{ji}/(1 - s_{0i})$: share among broadband adopters

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**Endogeneity:** high unobserved quality $\xi_{ji}$ raises both $p_{ji}$ and $\ln s_{j|\text{in},i} \Rightarrow$ two endogenous regressors

**Instruments**

- County and provider fixed effects absorb local demand and brand quality
- Rivals' speed differentiation, within and across tiers (Berry et al., 1995 ECTA): excluded from $j$'s utility, shift price and within-nest share through competition

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**Key assumption** (speeds chosen before $\xi$ is realized): $\mathbb{E}\left[\xi_{ji} \mid X_i, Z_i, \text{fixed effects}\right] = 0$

**2SLS** with county and provider fixed effects; strong first stages (Sanderson–Windmeijer $F$: 58.3 and 544.2)

▸ D2: Instrument formulas ▸ D3: Instrument strength ▸ D4: Hausman IVs

# Demand estimates

**Estimates**

- Price sensitivity: $\hat{\alpha} = -1.49$ per \$100
- Product similarity: $\hat{\rho} = 0.86$, so broadband products are close substitutes
- Mean own-price elasticity: $-4.99$

$\hat{\alpha} < 0$ and $\hat{\rho} \in [0, 1)$, as the model requires

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**Own-price elasticities in the literature**

- Kearns (2026, working paper), U.S. broadband: $-4.73$, $-4.46$, $-4.16$ for low, middle, high income
- Goetz (2019, working paper), U.S. TV providers: $-5.9$
- Hidalgo (2023, working paper), Colombia: $-4.6$
- Crawford et al. (2012, AER), cable and satellite: $-4.1$ and $-5.4$

▸ D5: Substitution ▸ D6: Distributions ▸ D7: Specifications

# Pricing reveals markups

**Pricing:** provider $f$ sets the prices of its products $\mathcal{J}_i^f$ in tract $i$ (Bertrand-Nash)

$$\max_{\{p_{ji}\}_{j \in \mathcal{J}_i^f}} \sum_{j \in \mathcal{J}_i^f} \left(p_{ji} - mc_{ji}\right) M_i \, s_{ji}\left(p_i, X_i, \xi_i\right) \tag{2}$$

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**Inversion:** stack the first-order conditions and solve for markups

$$p_i - mc_i = -\left(\Omega_i \odot \Lambda_i\right)^{-1} s_i$$

- $\Lambda_{jk,i} = \partial s_{ki} / \partial p_{ji}$, computed from $(\hat{\alpha}, \hat{\rho})$
- $\Omega_{jk,i} = 1$ if $j$ and $k$ belong to the same provider: multiproduct providers internalize cannibalization
- No cost data needed: marginal costs follow as $\widehat{mc}_{ji} = p_{ji} -$ markup

# Marginal costs and markups

**Cost function:** project recovered marginal costs on product characteristics

$$\log\left(mc_{ji}\right) = X_{ji}^{T}\gamma + \text{fixed effects} + \eta_{ji} \tag{3}$$

Imputes costs for products not offered, needed for Stage 1 portfolio choice

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**Results**

- Median Lerner index $L_{ji} = (p_{ji} - \widehat{mc}_{ji})/p_{ji}$: 0.20
- Median markup: \$14.3 per month
- Median marginal cost: \$57.5 per month

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**Do these magnitudes make sense?**

- Single-product benchmark: $1/|\bar{\varepsilon}| = 1/4.99 \approx 0.20$, the median Lerner index (back of the envelop)
- Implied price $\$57.5 + \$14.3 \approx \$72$/month vs. \$71 average U.S. internet bill (Parks Associates, 2024)

▸ D10: Cost estimates ▸ D11: Negative mc ▸ D9: Markups ▸ D6: Distributions

# First stage: Model of product positioning and fixed costs

# Product positioning: setup and challenge

**Setup**

- Firms know the second-stage model and the distribution of $(\xi, \eta)$, not their draws
- Offering product $j$ costs a fixed cost $FC_{ji}$
- Firms choose portfolios to maximize expected profits net of fixed costs

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**Approach: partial identification**

- Undominated choices (Fan and Yang, 2025, JPE) $\Longrightarrow$ moment inequalities
- Inference (Andrews and Soares, 2010, ECTA) $\Longrightarrow$ confidence set

Skip to estimates

# Observed product offerings reveal fixed costs

- $FC_{ji}$: fixed cost of offering product $j$ in tract $i$ (known to firms, not to me)
- $\Delta_{ji}$: expected incremental profit from offering product $j$, holding the firm's other choices fixed

**Offering rule:** for an observed offering, $\Delta_{ji} \geq FC_{ji}$

**Problem:** $\Delta_{ji}$ depends on rivals' choices, and equilibria are not unique $\Rightarrow$ bound it (Fan and Yang, 2025 JPE)

$$\underbrace{\Delta_{ji}^{\text{lower}}}_{\text{rivals offer everything}} \leq \Delta_{ji} \leq \underbrace{\Delta_{ji}^{\text{upper}}}_{\text{rivals offer nothing}}$$

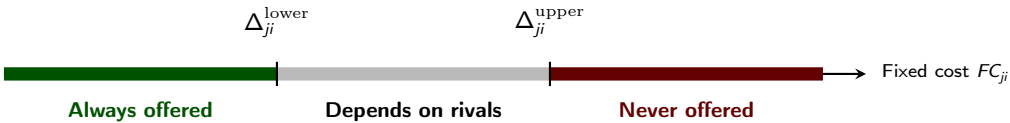
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▸ E1: Entry game

# From profit bounds to offer probabilities

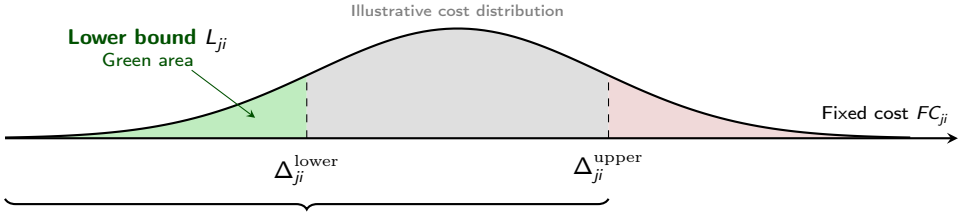
$$FC_{ji} = \mathrm{FC}_{ji}^{C} + \sigma_m \zeta_{ji}, \qquad \zeta_{ji} \sim \mathcal{N}(0,1) \tag{4}$$

$\mathrm{FC}_{ji}^{C}(\theta) = \theta_m + \theta_H 1\{j = H\}$; $m$: market-size tertile; $\sigma_m$: dispersion; $F_\zeta$: standard normal CDF

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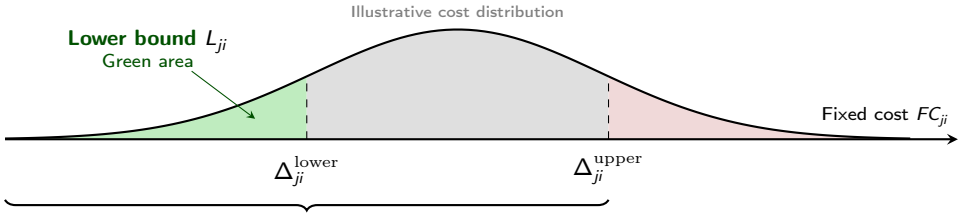
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$\mathrm{FC}_{ji}^{C}(\theta) = \theta_m + \theta_H 1\{j = H\}$; $m$: market-size tertile; $\sigma_m$: dispersion; $F_\zeta$: standard normal CDF




$$F_\zeta\left(\frac{\Delta_{ji}^{\text{lower}} - \mathrm{FC}_{ji}^{C}(\theta)}{\sigma_m}\right) \le \Pr(Y_{ji} = 1 \mid X_{ji}) \le F_\zeta\left(\frac{\Delta_{ji}^{\text{upper}} - \mathrm{FC}_{ji}^{C}(\theta)}{\sigma_m}\right). \tag{5}$$

▸ E2: Bound diagnostics

# Moment inequalities for fixed costs

**Bounds on offer probabilities $\Rightarrow$ two moment functions**

$$m_{ji}^{L}(\theta, \sigma_m) = F_{\zeta}\left(\frac{\Delta_{ji}^{\text{lower}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right) - 1(Y_{ji} = 1), \qquad m_{ji}^{U}(\theta, \sigma_m) = 1(Y_{ji} = 1) - F_{\zeta}\left(\frac{\Delta_{ji}^{\text{upper}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right) \tag{6}$$

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**Interact with non-negative instruments** $g^{(k)}(X_{ji}, W_i)$, $k = 1, \ldots, K$ ($K = 33$: 66 inequalities), and stack:

$$m(\theta,\sigma_m) = \frac{1}{|\mathcal{J}_i|}\sum_{j\in\mathcal{J}_i}\begin{pmatrix} m_{ji}^{L}(\theta,\sigma_m)\, g^{(k)}(X_{ji}, W_i) \\ m_{ji}^{U}(\theta,\sigma_m)\, g^{(k)}(X_{ji}, W_i) \end{pmatrix}_{k=1,\ldots,K} \in \mathbb{R}^{2K} \tag{7}$$

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**Identified set**: all parameters consistent with every inequality

$$\Theta_0 = \{(\theta,\sigma_m) \in \Theta : \ \mathbb{E}\left[m(\theta,\sigma_m)\right] \leq 0\} \tag{8}$$

# Inference on fixed-cost parameters

**Test inversion** (Andrews and Soares, 2010, ECTA)

- For each candidate $(\theta, \sigma_m)$, test

$$H_0 : \quad \mathbb{E}[m(\theta, \sigma_m)] \leq 0$$

▸ E3: Parameter bounds

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**Search over 7 parameters** (a grid is infeasible)

1. Minimize $T_n$ to obtain a starting point.

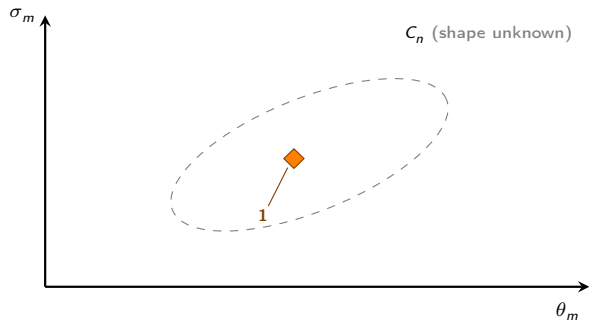



Illustration: two of the seven parameters

▸ E3: Parameter bounds

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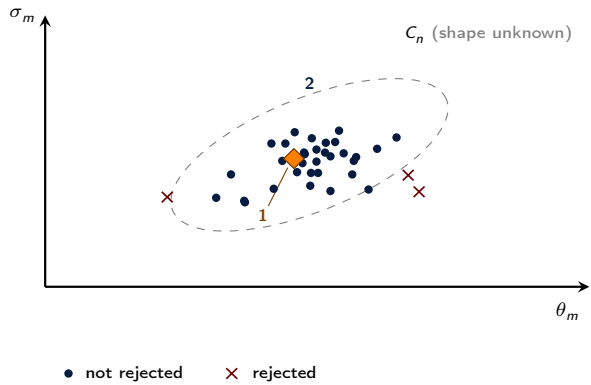



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**Search over 7 parameters** (a grid is infeasible)

1. Minimize $T_n$ to obtain a starting point.
2. Run random-walk chains; keep draws that are not rejected.
3. Explore further from the edge of the accepted region.

$\Rightarrow$ **656 accepted parameter vectors**

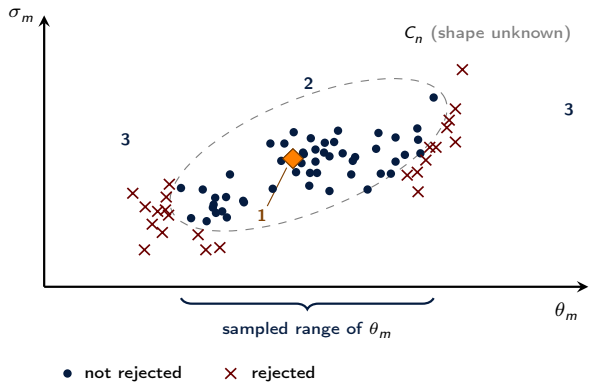



Illustration: two of the seven parameters

▸ E3: Parameter bounds

# Fixed costs are large and only partially identified

| | 95% Confidence Set ($/month) |
|---|---|
| *Mean fixed cost: low-speed position* | |
| Small market | [7,357; 74,287] |
| Medium market | [5,155; 84,040] |
| Large market | [5,832; 82,783] |
| *High-speed fixed-cost differential* | |
| High vs. low speed | [−53,843; 25,081] |
| *Fixed-cost heterogeneity* | |
| Small market | [0; 41,542] |
| Medium market | [0; 45,221] |
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- Wide sets reflect **partial identification**
- Mean fixed cost per product: **about $60k–$1M per year**
- Sets overlap across market sizes; dispersion and the high-speed premium are not pinned down
- Same order of magnitude as sunk deployment costs in Kearns (2026, working paper)

▸ E3: Bounds ▸ E4: Plausibility

# Counterfactuals

# Counterfactuals: affordability versus deployment

Compare each policy with the same **no-subsidy baseline**

## ACP: affordability

- Targets **low-income households**
- Up to **$30/month** off broadband service
- Discounts delivered through **participating providers**

**In the model:** $30 off eligible households' bills; **firms reoptimize prices** and are reimbursed, as under the program

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## BEAD: deployment

- Targets **unserved and underserved areas**
- **NTIA** allocates funds to states
- States award **deployment grants** to providers
- Typically up to **75% BEAD funding**, with at least **25% matching funds**

**In the model:** 25–75% of fixed costs paid; providers re-choose products, then prices

▸ F1: BEAD rules ▸ F2: Accounting

# ACP buys affordability; BEAD buys infrastructure

**ACP at $30/month**

| | |
|---|---|
| $\Delta$CS per month | $+$\$0.80–0.82B |
| Public cost | \$0.81–0.83B/month |
| Discount on gross price | $-$\$30 eligible ($-$\$9 avg.) |
| Take-up change | $+ \sim$1.5pp |
| New firm-market entries | 10–17 |

**Mechanism:** lower household bills $\Longrightarrow$ higher adoption

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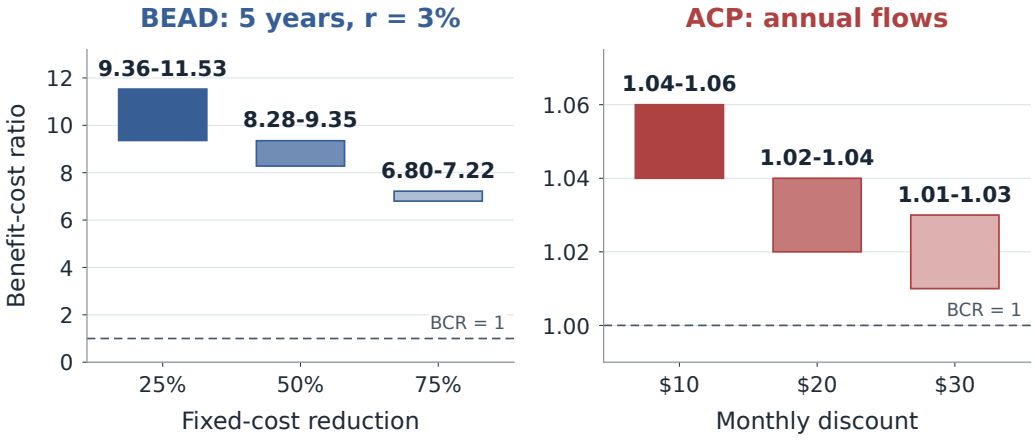
### BEAD at 75% fixed-cost subsidy

| | |
|---|---|
| ΔCS per month | +$1.43–1.84B |
| Public cost | $14.3–17.9B one-time |
| Take-up change | + ∼9pp (73% → 82%) |
| New firm-market entries | 6,665–8,648 |
| Newly served tracts | 6,128–8,054 |

**Mechanism:** lower fixed costs $\Longrightarrow$ entry $\Longrightarrow$ coverage

▸ F3: ACP detail ▸ F4: BEAD detail ▸ F5: Repricing

# BEAD: $7 per dollar in five years. ACP: near break-even




Welfare = ΔCS+ΔPS. BEAD: five-year PV ($r = 3\%$) / one-time grant; per year, about $1.5 per grant dollar. ACP: annual welfare / annual outlay, paid every year.

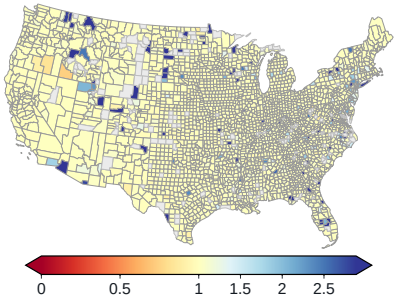


▸ F7: Sensitivity ▸ F2: Accounting ▸ F8: Sequencing detail

# Welfare per public dollar varies across counties

**ACP: affordability subsidy**

$30/month discount: about **$1** per public dollar



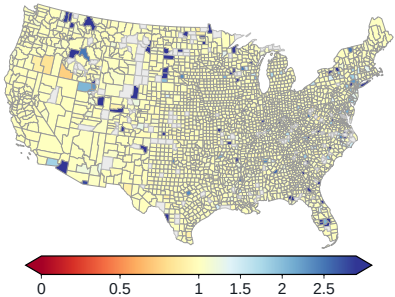

**Annual welfare gain / annual spending**

$9.9–10.1B / $9.7–9.9B per year

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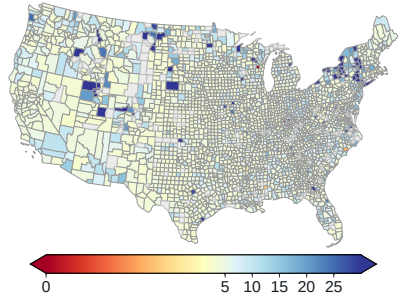
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**Annual welfare gain / annual spending**
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## BEAD: deployment subsidy

75% subsidy: about **$7** per public dollar




**5-year PV of welfare / one-time grant**
$99–129B / $14.3–17.9B ($r = 3\%$)

Colour scales differ across maps. BEAD's cost is treated as a one-time capital grant; ACP's as a recurring transfer.

▸ Returns by subsidy level ▸ F7: Sensitivity

# BEAD gains track market structure; ACP gains track income

**Consumer-surplus gains by market and income type**

| Consumer-surplus gain ($/hh/month) | BEAD (75%) |
|---|---|
| Monopoly tracts | +41.5 to +51.6 |
| Duopoly tracts | +2.0 to +3.1 |
| Lowest-income quartile | +16.7 to +20.7 |
| Highest-income quartile | +9.9 to +14.4 |

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**Consumer-surplus gains by market and income type**

| Consumer-surplus gain ($/hh/month) | BEAD (75%) | ACP ($30) |
|---|---|---|
| Monopoly tracts | +41.5 to +51.6 | +3.2 to +3.5 |
| Duopoly tracts | +2.0 to +3.1 | +6.0 to +6.1 |
| Lowest-income quartile | +16.7 to +20.7 | +8.4 to +8.6 |
| Highest-income quartile | +9.9 to +14.4 | +2.2 to +2.4 |

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| Lowest-income quartile | +16.7 to +20.7 | +8.4 to +8.6 |
| Highest-income quartile | +9.9 to +14.4 | +2.2 to +2.4 |

BEAD gains follow market structure (monopoly $\gg$ duopoly); ACP gains follow income (lowest quartile $> 3\times$ highest)

▸ F9: Full table

# Conclusion

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- **Question:** with one public dollar, lower household bills (ACP) or help firms build networks (BEAD)?
- **Approach:** structural model of household demand, provider pricing, and product offerings, with fixed costs bounded by moment inequalities
- **Findings** (within five years)
  - BEAD returns about **$7 per public dollar**, concentrated in unserved and monopoly tracts
  - ACP returns about **$1 per public dollar**: largely a transfer
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  - BEAD gains track market structure; ACP gains track income
- **Policy implication:** the gains from each instrument depend on the local barrier to access

▸ G2: Limitations

# Thank you

**Sidi Sawadogo**

Department of Economics and CIREQ
Université de Montréal

sidi.mohamed.sawadogo@umontreal.ca • sidi-sawadogo.github.io

**Read the paper**

*Build or Subsidize? The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband*

Scan the QR code or click here to access the paper.

# Backup Slides


- A. Introduction
- B. Data
- C. Reduced-form evidence
- D. Demand and pricing
- E. Entry and fixed costs
- F. Counterfactuals
- G. Conclusion

# Backup A1: Program timeline and implementation

- **Feb–Dec 2021**: Emergency Broadband Benefit (EBB), ACP's predecessor, up to $50/month, nearly 9 million households
- **Nov 2021**: IIJA enacted; $14.2B ACP, $42.45B BEAD
- **Jan 2022**: ACP launch; up to $30/month ($75 Tribal), one-time $100 device discount; peak 23 million households
- **Jun 2023**: NTIA announces BEAD state allocations; state proposals due Dec 2023
- **Jun 2024**: ACP funding exhausted; Congress declines to reappropriate
- **2024–2026+**: BEAD subgrantee selection, permitting, construction

ACP's full cycle (launch to lapse) sits inside the 2014–2025 event window; BEAD's does not, which is why its steady state comes from the structural counterfactuals.

Back

# Backup B1: Data sources

| Source | What it contributes | Used for |
|---|---|---|
| FCC Urban Rate Survey | product prices, speed tiers, technology | Event studies, demand prices |
| FCC Form 477 | Provider deployment at block/tract level | Product availability, feasible entry set |
| ACS | Income, poverty, household counts | ACP exposure, market size, controls |
| FCC adoption data | Tract-level adoption bins | Outside option, market shares |
| NTIA records | BEAD allocations by state | BEAD treatment intensity |

- Event-study panel: 52 states (50 states, DC, Puerto Rico), 2014–2025, $N = 624$ cells
- Demand sample: 170,419 observed products in 66,454 tracts
- Entry sample: 727,601 potential positions in 70,854 tracts

Back

# Backup B2: Market share construction and validation

- Product-level subscriber counts are never disclosed; shares are **inferred**, following the logic of Crawford et al. (2019)
- FCC tract-level adoption bins anchor total adoption and the outside option; logit weights built from product characteristics and demographics allocate the total across available products
- These are inferences, not observed counts, and are treated as such
- **Validation**: the construction implies 93.7M subscribing households at $\geq$10/1 Mbps, 99.2% of the FCC's independent 94.5M benchmark
- Mean outside-option share: 19.1%; mean product share 0.32; provider HHI computed from within-tract subscriber shares

Back

# Backup B3: Form 477 coverage overstatement

- Known limitation: a block is marked served if a provider *could* serve a household there; this overstates availability in multi-dwelling buildings and rural blocks
- I use the **December 2020** release, before the post-IIJA transition to the Broadband Data Collection system, so the overstatement is **uniform across the cross-section**
- It therefore does not differentially affect the treatment variation used for identification
- Provider normalization (1,732 canonical identifiers) prevents overstating competition by counting subsidiaries separately

Back

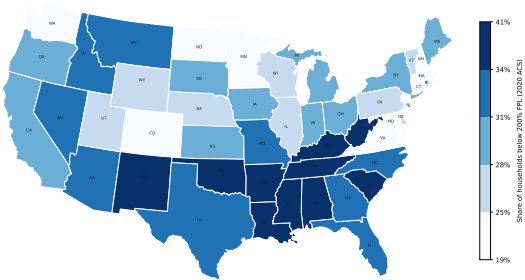
# Backup B4: Sample construction

- No single federal source links prices, deployment, demographics, and adoption; five sources are merged at census-tract resolution
- **Brand-first price matching** assigns a price to each of the 727,601 potential positions (provider-tier-tract triples)
- **Provider normalization**: names manually mapped to 1,732 canonical corporate identifiers, so subsidiaries are not counted as separate competitors
- Demand sample (66,454 tracts) is smaller than the entry sample (70,854) because 4,400 tracts with observed products did not match tract-level covariates
- Prices deflated to constant dollars; a product is a provider-speed-tier pair, $g \in \{H, L\}$ split at 10/1 Mbps

Back

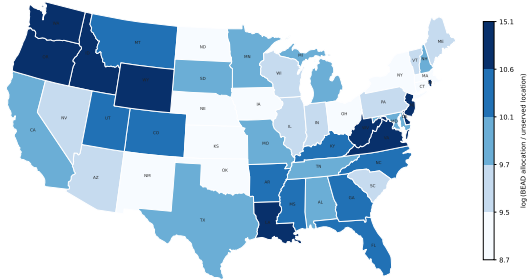
# Backup C1: Exposure maps

**ACP exposure**




Share below 200% FPL (2020 ACS)

**BEAD exposure**




Log allocation per unserved location

The two exposures are far from collinear: high-poverty states (South) and high-cost-per-location states (Mountain West, Pacific Northwest) barely overlap.

Back

# Backup C2: Event-study specification detail

- Outcomes (six, three margins): log average real price; log price by tier (below/above 25/3 Mbps); product shares by tier; provider HHI
- Prices are product-count-weighted state-year averages, deflated by CPI-U
- Event-time normalization: $\bar{t}^{ACP} = 2022$ (first full implementation year); $\bar{t}^{BEAD} = 2021$ (enactment); omitted period $\tau = -1$
- Five pre-treatment leads test parallel trends; regressions weighted by product counts; SEs clustered at the state level
- **State-specific linear trends**: high-poverty and high-unserved states may sit on different broadband trajectories; trends absorb linear deviations, so identification comes from non-linear departures coinciding with rollout

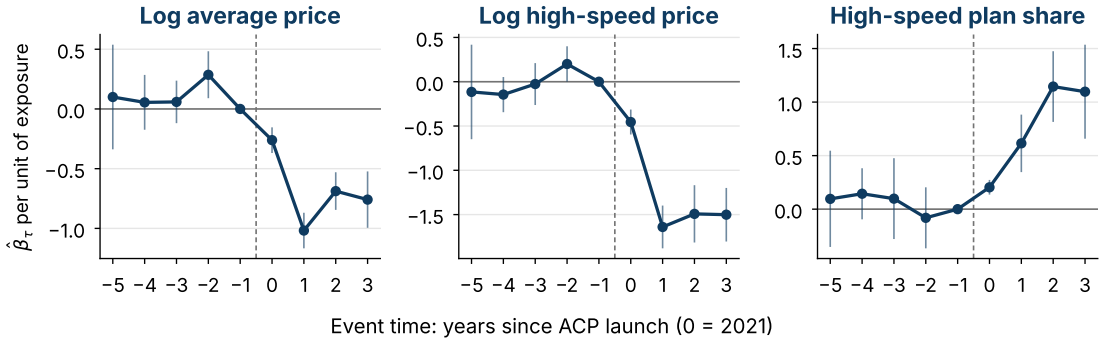
Back

# Backup C3: Anticipation and policy endogeneity

- **Policy endogeneity**: both intensities are predetermined; 2020 ACS demographics for ACP, a federal formula on pre-grant unserved counts for BEAD; neither responds to post-IIJA ISP behavior
- **BEAD anticipation**: announced Nov 2021, allocations finalized 2023; anticipatory responses would appear at $\tau = 1$–$2$, where estimates are near zero
- **ACP anticipation**: the EBB ran Feb–Dec 2021, so the $\tau = -1$ base period may absorb some anticipatory ISP behavior; this **understates** ACP's total effect
- Neither concern drives the qualitative results

Back

# Backup C4: ACP headline effects, scaled to P25 vs. P75 exposure




**Effects, P25 vs. P75 exposure**

- Prices: −7 to −10% overall, −15 to −18% for high-speed plans
- High-speed plan share: +6 to +11pp

▸ C5: Pooled DiD ▸ Back

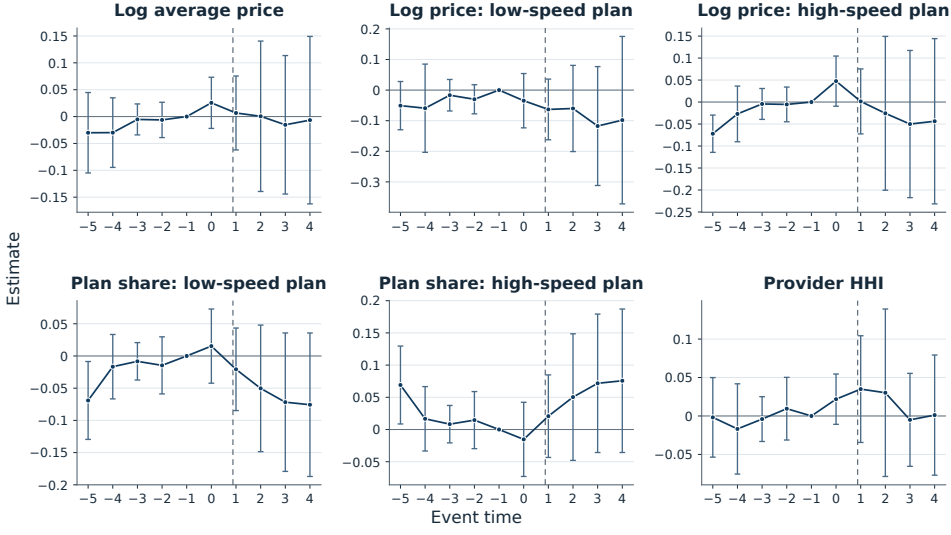
# Backup C5: Pooled difference-in-differences estimates

| Outcome | ACP (FPL200) | BEAD (log alloc.) |
|---|---|---|
| Log average price | −0.561 (sig.) | small, n.s. |
| Log high-speed price | −0.922 (sig.) | small, n.s. |
| High-speed product share | +0.390 (sig.) | small, n.s. |
| Provider HHI | −0.306 (sig.) | small, n.s. |

- Pooled post-period TWFE estimates confirm the event-study dynamics for ACP
- BEAD's pooled estimates are small and insignificant, consistent with effects that emerge only late in the window and that the urban URS frame attenuates
- Flat product-composition results for BEAD are consistent with entry, not incumbent repositioning, as its operative channel

Back to ACP Back to BEAD

# Backup C6: BEAD event-study estimates



# Backup C7: The URS urban frame

- The FCC Urban Rate Survey samples providers in **urban tracts**; the event-study panel under-represents exactly the rural markets BEAD targets
- If BEAD's price and concentration effects concentrate in rural areas, as the structural results imply, URS-based estimates understate its impact: a **conservative lower bound**
- ACP works through demand expansion in already-served, largely urban markets, so the URS frame captures its effects more directly
- This asymmetry is one reason the counterfactuals, not the event studies, carry BEAD's welfare quantification

Back

# Backup D1: Why nested logit rather than random coefficients

- Random coefficients allow richer substitution but can generate **non-unique pricing equilibria** in multiproduct settings (Nocke and Schutz, 2018): an unstable mapping from primitives to prices
- The counterfactuals **re-solve prices at every iteration** of the positioning game, across 70,854 tracts and 50 draws; a unique closed-form equilibrium is essential
- The single nest applies uniformly: a speed-tier nest is degenerate wherever only one tier is offered (many rural tracts)
- Restrictiveness check: elasticity $-4.99$ and Lerner 0.20 sit inside the ranges of Kearns (2026, working paper), Goetz (2019), and Crawford et al. (2019)
- Elasticity heterogeneity is still rich through shares: $\approx -2.1$ at $s_{jt} = 0.10$ (rural, thin markets) vs. $\approx -8.6$ at $s_{jt} = 0.45$ (dense markets); ACP gains concentrate where elasticity is high, BEAD gains where fixed costs bind

Back

# Backup D2: Differentiation instruments

Four instruments, for each speed characteristic $x \in \{\text{download},\ \text{upload}\}$ (Berry et al., 1995 ECTA):

$$Z_{ji}^{\text{same tier}}(x) = \sum_{\substack{k \neq j \\ \text{same tier}}} (x_{ji} - x_{ki})^2, \qquad Z_{ji}^{\text{other tier}}(x) = \sum_{\substack{k \\ \text{other tier}}} (x_{ji} - x_{ki})^2$$

- Measure how differentiated product $j$ is from rivals' products in the same tract, within and across speed tiers
- Rivals' speeds shift $j$'s price and within-nest share through competition, not through $j$'s utility
- Two endogenous regressors ($p_{ji}$, $\ln s_{j|\text{in},i}$), four excluded instruments

▸ D3: Instrument strength ▸ Back

# Backup D3: Instrument strength

- Four Berry et al.(1995, ECTA) quadratic differentiation instruments: within- and cross-tier characteristic dispersion in download and upload speed
- Economically transparent signs: more within-tier dispersion raises own price (slack competitive pressure); more cross-tier dispersion lowers the within-nest share
- Sanderson-Windmeijer conditional $F$: **58.3** (price) and **544.2** (within-share), far above the Stock-Yogo 10% critical value of 7.56
- Cragg-Donald minimum eigenvalue: **986.2**
- Shea partial $R^2$ for price is 0.0054: the instruments explain a small share of residual price variation, but the identified variation is clean; see D4 for the endogenous alternative

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# Backup D4: Alternative instruments (Hausman-style)

- An alternative specification uses CMT-style Hausman instruments (Ciliberto, Murry and Tamer, 2021): same product's price in other counties
- Stronger first stage on price (Shea $R^2 = 0.393$) but $\hat{\alpha} = -0.123$, close to OLS with fixed effects, and **99.9% of products imply negative marginal costs**
- Convergence to OLS indicates the Hausman instrument is endogenous conditional on two-way fixed effects, likely because national providers carry correlated brand quality across counties
- The (Berry et al., 1995 ECTA) specification is preferred on supply-side plausibility grounds

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# Backup D5: Substitution patterns implied by the nest

**Elasticities in closed form from $(\hat{\alpha}, \hat{\rho})$ and observed shares**

$$\text{Own price:} \quad \varepsilon_{jj,i} = \hat{\alpha}\, p_{ji} \left[ \frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{j|\text{in},i} - s_{ji} \right]$$

$$\text{Cross price } (k \neq j): \quad \varepsilon_{jk,i} = -\hat{\alpha}\, p_{ki} \left[ \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{k|\text{in},i} + s_{ki} \right]$$

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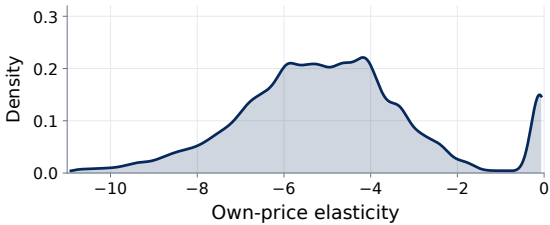
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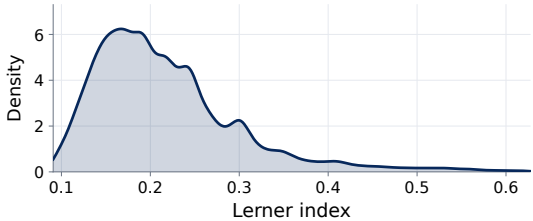
**Reading the elasticities**

- The within-nest term $\hat{\rho}/(1-\hat{\rho}) \cdot s_{k|\text{in},i}$ captures substitution among broadband products
- With $\hat{\rho} = 0.86$, it carries a weight of about 6: a price cut draws mostly from rival broadband products, much less from non-adopters
- Separates **business stealing** among providers from **new adoption**

▸ D1: Why one nest ▸ Back

# Backup D6: Distributions of elasticities and Lerner indices




**Mean own-price elasticity:** $-4.99$

**Median Lerner index:** 20%

$L_{jt} = (p_{jt} - mc_{jt})/p_{jt}$

Single-product benchmark:

$1/4.99 \approx 20\%$

▸ D8: Demand details ▸ D9: Markups ▸ Back

# Backup D7: Estimation procedure and specifications

**Procedure**

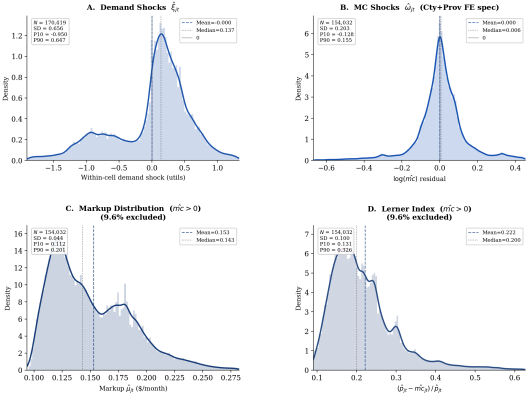
1. Build $\ln s_{ji} - \ln s_{0i}$ and $\ln s_{j|\text{in},i}$ from observed shares
2. Remove county and provider fixed effects (within transformation)
3. 2SLS: instrument $p_{ji}$ and $\ln s_{j|\text{in},i}$ with $Z_{ji}$; speeds and their squares as controls

| | (1) OLS | (2) OLS + fixed effects | (3) 2SLS | (4) 2SLS + fixed effects |
|---|---|---|---|---|
| Price/\$100 [$\hat{\alpha}$] | $-0.344$ | $-0.127$ | $+3.261$ | $-1.491$ |
| Within-share [$\hat{\rho}$] | 1.183 | 1.029 | 1.242 | 0.863 |
| Mean own-price elasticity | $+0.83$ | $+1.96$ | $-5.91$ | $-4.99$ |
| County + provider fixed effects | No | Yes | No | Yes |

- Only (4) gives $\hat{\alpha} < 0$ and $\hat{\rho} \in [0, 1)$: without IVs, $\hat{\rho} > 1$; without fixed effects, the price coefficient flips sign
- $N = 170{,}419$; state-level aggregation gives elasticity $-5.87$

▸ D3: Instrument strength ▸ Back

# Backup D8: Demand and supply diagnostics




- Demand shocks $\hat{\xi}_{jt}$ centered near zero: instruments orthogonal to the demand residual after county and provider absorption
- Marginal-cost shocks likewise symmetric
- Markup distribution right-skewed, driven by multiproduct firms
- Lerner median 0.200 matches the inverse-elasticity benchmark $1/4.99$

Back

# Backup D9: Markups and Lerner indices by tier and firm type

| | All | Tier H | Tier L |
|---|---|---|---|
| Median markup ($/month) | 14.3 | 14.2 | 15.3 |
| Median marginal cost ($/month) | 57.5 | 58.9 | 43.0 |
| Median Lerner index | 0.200 | 0.195 | 0.278 |
| | | Single-product | Multi-product |
| Median Lerner index | | 0.191 | 0.278 |
| Mean Lerner index | | 0.204 | 0.322 |

- Single-product mean Lerner 0.204 $\approx 1/|\bar{\varepsilon}| = 0.200$: the inverse-elasticity check passes
- Tier L (legacy DSL) marks up more: de facto monopolists in low-density areas
- Multiproduct firms internalize cannibalization and price both tiers higher: a new rival disciplines prices in a way an incumbent's second tier does not, the margin on which BEAD operates

Back

# Backup D10: Marginal-cost estimates

- Recovered costs regressed on characteristics: $\ln(\widehat{mc}_{jt}) = W'_{jt}\gamma + \omega_{jt}$, specification without fixed effects preferred because cost variation runs across providers, technologies, and tiers
- Baseline variable cost $\approx$ **\$33/month** at zero measured speeds; Tier H premium $\approx$ **70%**
- Mean marginal cost $\approx$ \$59.5/month; Tier H \$61.5 vs. Tier L \$38.4: faster connections require heavier productt
- Higher upload speeds correlate with *lower* costs, consistent with fiber-grade networks combining upload capacity with newer, cost-efficient productt
- Download speed raises cost at a diminishing rate
- Column (1) imputes marginal costs for all 727,601 potential positions in the entry sample

Back

# Backup D11: Negative marginal-cost diagnostics

- 16,387 of 170,419 products (**9.62%**) imply negative marginal costs
- A known artifact of Bertrand-Nash markup recovery: cannibalization-adjusted markups can exceed price when substitution within a firm's own portfolio is strong
- Negative-mc products cluster among **multiproduct firms** (13.5% vs. 8.9% for single-product firms)
- Cost regressions and entry-stage imputation use the positive-mc subsample (90.4%)
- The preferred specification minimizes the problem: the Hausman alternative (D4) produces 99.9% negative costs and is rejected on exactly these grounds

Back

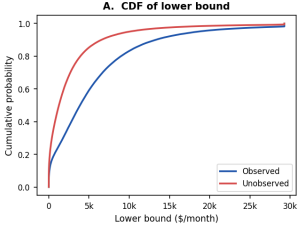
# Backup E1: The entry game

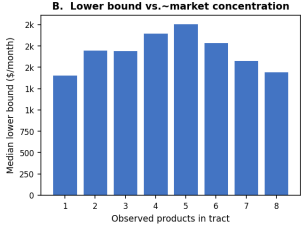
$$\max_{Y_t^f \in \mathcal{Y}_t^f} \mathbb{E}_{\xi,\eta}\left[r_t^f(Y_t^f, Y_t^{-f}) \mid \mathcal{I}^f\right] - \sum_{j \in \mathcal{J}_t^f} Y_{jt}\left(\theta_{b(t)} + \theta_H 1\{g = H\} + \sigma_{b(t)}\zeta_{jt}\right) \tag{9}$$

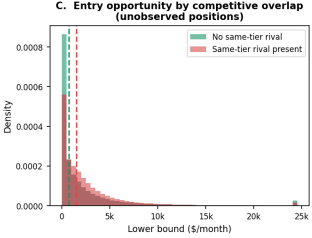
- Firms choose product portfolios before demand and cost shocks realize; expected variable profits come from the estimated demand and pricing stage
- Multiple equilibria imply **bounds** rather than point estimates; no selection rule is imposed
- Moment inequalities compare observed choices with one-position deviations: drop an offered product, or add a feasible unoffered one
- Identification: cross-tertile offering rates discipline $\theta_b$; within-tertile H vs. L comparisons discipline $\theta_H$; the sharpness of the entry threshold disciplines $\sigma_b$
- BEAD enters as a proportional fixed-cost reduction

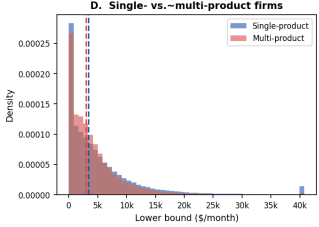
Back

# Backup E2: Profit-bound diagnostics




- Median tightness ratio $\Delta^{lower}/\Delta^{upper}$ is 0.017: competitive lower bounds are small relative to monopoly upper bounds
- Observed positions have systematically higher ratios: firms select into profitable opportunities
- All valid positions satisfy $\Delta^{lower} > 0$ and $\Delta^{lower} \leq \Delta^{upper}$: no dominated positions; the upper-bound restriction drives identification

Back

# Backup E3: Fixed-cost parameter bounds

| | 95% CS ($/month) |
|---|---|
| $\theta_H$ (high-speed premium) | [−53,843; 25,081] |
| $\theta_{\text{Small}}$ | [7,357; 74,287] |
| $\theta_{\text{Medium}}$ | [5,155; 84,040] |
| $\theta_{\text{Large}}$ | [5,832; 82,783] |
| $\sigma_{\text{Small}}$ | [0; 41,542] |
| $\sigma_{\text{Medium}}$ | [0; 45,221] |
| $\sigma_{\text{Large}}$ | [0; 37,891] |

- Coordinate-wise projections of the 95% Andrews-Soares confidence set; intercepts by market-size tertile
- Intervals reflect partial identification under multiple equilibria, not sampling noise
- Counterfactuals draw 50 parameter vectors from the set and report outcome bounds

Back

# Backup E4: Are the fixed-cost magnitudes plausible?

- Monthly intercept bounds of \$5,155–\$84,040 correspond to roughly **\$62k–\$1.0M per year** per product position: the flow equivalent of annualized capital and operating expenditure
- Kearns (2026, working paper) recovers sunk deployment costs of \$560k–\$770k per location for de novo entry (DSL to fiber) from revealed decisions in Seattle: the same order of magnitude
- Revealed-preference cost estimates exceed engineering figures because they embed opportunity costs and regulatory barriers; that applies equally here
- $\sigma_b$ lower bounds touch zero: a nearly deterministic fixed-cost structure is not ruled out
- $\theta_H$ straddles zero: with construction sunk, ongoing fixed costs of a fast tier need not exceed a slow tier's

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# Backup F1: BEAD allocation formula and eligibility

- A federal formula allocates BEAD funds to states by the count of **unserved and underserved locations**: broadband-serviceable locations with no service, service below 25/3 Mbps, or latency above 100 ms
- Counts are measured **pre-grant**: treatment intensity is fixed before any post-IIJA outcome
- BEAD can cover up to **75% of eligible project costs**, with a minimum 25% subgrantee match; this motivates the 25/50/75% counterfactual grid
- Exposure measure: log allocation per unserved location; higher values mean a more intensive supply-side intervention per targeted location

Back

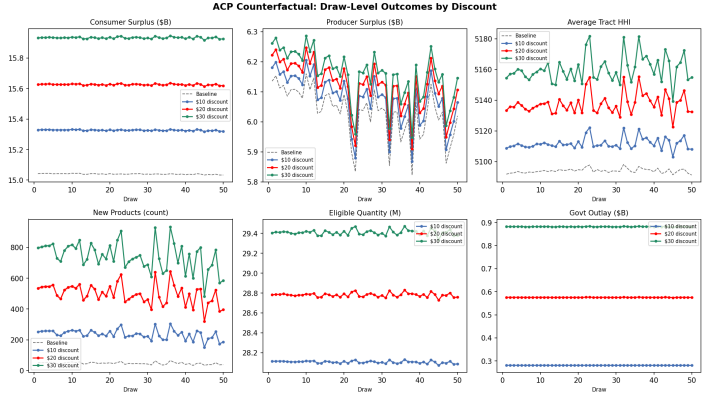
# Backup F2: Welfare accounting and CBA conventions

| | BEAD | ACP |
|---|---|---|
| Fiscal cost | One-time grant lowering private fixed costs | Recurring transfer on eligible subscriptions |
| Welfare gain | PV of annual $\Delta$CS + $\Delta$PS, $r = 3\%$, $T = 5$ | Annual flow $\Delta$CS + $\Delta$PS |
| Statistic | NPV and BCR | Annual net surplus and BCR |

- Benefits are the discounted surplus the model attributes to each program, **nothing else**: no spillovers into employment, schooling, or health, so ratios understate full social returns
- The model delivers an annual flow, not a time path, so any horizon is an assumption laid on top of it. A five-year window makes that assumption cheap and is the **most conservative** horizon in the sensitivity table
- The two BCRs are not directly comparable levels: one prices a capital stock, the other a flow; the object of interest is the **sequencing** they jointly imply
- Estimates exclude administrative and implementation costs on both sides

Back

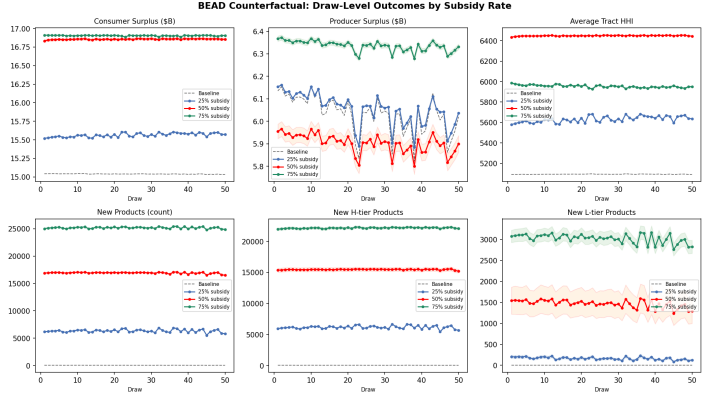
# Backup F3: ACP draw-level outcomes




ACP outcomes are stable across draws because the operative channel is demand rather than entry; fixed-cost uncertainty barely matters for a price discount.

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# Backup F4: BEAD draw-level outcomes




Across 50 fixed-cost draws, BEAD generates large new-product responses, especially at the 75% rate. Reported bounds are $[Q_{2.5}, Q_{97.5}]$ across draws combined with equilibrium multiplicity.

▸ Back

# Backup F5: What if gross prices respond to ACP?

- The ACP counterfactual holds gross posted prices fixed by design: it isolates the **demand-expansion channel** of the statutory discount
- The event study shows equilibrium repricing favors consumers: prices 7–10% lower in high-eligibility states after launch
- Allowing repricing would therefore **add** consumer gains; the structural CS estimates are conservative on this margin
- Read the counterfactual as the welfare effect of the discount plus induced demand and entry responses, not a decomposition of posted-price adjustment
- The firm's per-subscriber margin is unchanged because the government reimburses at the gross price

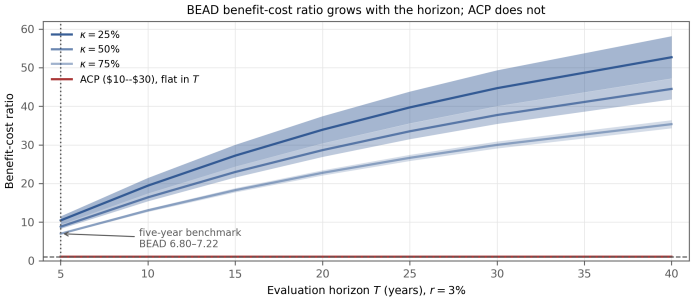
Back

# Backup F6: Why average HHI rises under BEAD

- At the 75% subsidy, average tract HHI moves from [4,925; 5,199] to [5,493; 5,603]
- Pure **composition effect**: BEAD serves 6,128–8,054 previously unserved tracts, and a tract served by one firm has an HHI of 10,000
- Adding newly served monopoly tracts raises the cross-sectional mean even as concentration falls *within* already-served tracts
- Consumer surplus in each newly served tract rises from zero, so welfare rises despite the index
- In this context the index measures expanded access, not increased market power

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# Backup F7: The BEAD ratio scales with the horizon




- The cost is paid once and is not discounted; only the welfare stream is, so the BCR **scales with the annuity factor**
- Across rates 1–7% at five years, every BEAD lower bound stays **above 6** and NPV stays positive ($74.6–$118.2B at 75%)
- The *sign* is robust; the *magnitude* is not. Quote the horizon with the number
- ACP's ratio is invariant to $r$ and $T$; any non-trivial infra-marginal share pushes its efficiency-adjusted BCR below one

| Horizon $T$ (years), $\kappa = 75\%$, $r = 3\%$ | 5 (benchmark) | 10 | 15 | 25 | 40 |
|---|---|---|---|---|---|
| BEAD BCR | 6.80–7.22 | 12.67–13.44 | 17.73–18.81 | 25.85–27.44 | 34.32–36.42 |

# Backup F8: The sequencing logic, and one honest gap

- Unserved, high-cost markets: infrastructure absence binds; fixed-cost relief delivers large, durable gains *Model implication*
- In served low-income markets, the binding constraint is price: an affordability program raises take-up and surplus at roughly break-even fiscal efficiency *Model implication*
- Where BEAD induces entry, it creates the supply-side conditions (more competing products) that raise the efficiency of a subsequent demand subsidy
- Reconciles with Kearns (2026, working paper): in already-served Seattle the demand-side tool wins, exactly the condition under which this framework also favors it

## One honest gap, and it is BEAD-specific

Duopoly tracts gain only \$2.0–\$3.1 per household from BEAD: once one competitor exists, a fixed-cost discount adds little, and a subsidized third entrant needs more than 75%. ACP does reach these tracts (\$6.0–\$6.1, its largest stratum), so the two instruments are not both weak in the same place.

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# Backup F9: Distributional effects, full table

| Market type | Baseline CS [$/hh/mo] | BEAD ΔCS [$/hh/mo] | ACP ΔCS [$/hh/mo] |
|---|---|---|---|
| *By median household income* | | | |
| Q1 (lowest, ≤$44k) | 87.5–89.9 | +16.7–20.7 | +8.4–8.6 |
| Q2 ($44k–$59k) | 109.0–111.8 | +12.4–15.4 | +5.9–6.2 |
| Q3 ($59k–$81k) | 124.7–127.9 | +9.8–13.0 | +4.1–4.3 |
| Q4 (highest, >$81k) | 133.9–138.5 | +9.9–14.4 | +2.2–2.4 |
| *By number of active providers* | | | |
| Monopoly | 76.5–86.5 | +41.5–51.6 | +3.2–3.5 |
| Duopoly | 132.4–133.3 | +2.0–3.1 | +6.0–6.1 |
| 3+ providers | 124.8–125.9 | +2.5–3.8 | +4.9–5.2 |
| *By tract population* | | | |
| Q1 (lowest, ≤2,953) | 99.9–105.5 | +10.7–19.4 | +6.9–7.2 |
| Q4 (highest, >5,581) | 124.9–128.3 | +9.7–12.9 | +2.8–3.0 |

75% BEAD subsidy; $30 ACP discount. $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws, differenced draw-by-draw against the same draw's baseline. Monopoly tracts start from the *lowest* baseline CS and gain the most from BEAD. Panel B conditions on $\geq 1$ baseline provider, so it excludes the wholly unserved tracts BEAD also reaches.

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# Backup G1: External validity and scope

- Estimates are long-run structural returns conditional on the modeled entry response; they exclude administrative and implementation costs
- The 2020 baseline predates both programs; post-2020 cost shocks (construction inflation, labor) would shift levels but not the ordering of margins
- The complementarity is a property of the deployment technology (costly to build, cheap to use), not of these two programs: the same margins recur in electricity, water, transportation, and payments
- The Seattle contrast (Kearns, 2026) is the right external check: where deployment gaps are closed, the demand-side tool wins, exactly as this framework predicts

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# Backup G2: Limitations in full

**Measurement and data**

- URS prices are urban: BEAD reduced-form effects are lower bounds; rural price levels extrapolated
- Market shares inferred from adoption bins, not observed; validated nationally within 1%
- 9.6% negative marginal costs; excluded and documented

**Model and extrapolation**

- ACP counterfactual holds gross prices fixed: a lower bound
- Wide fixed-cost bounds: partial identification by design
- The BEAD BCR scales with the horizon (6.80–7.22 at five years; 25.85–27.44 at 25); the sign never changes
- Spillovers and administrative costs excluded on both sides

None of these reverses the ordering: BEAD's five-year floor of 6.80 and ACP's near-break-even hold at every rate and horizon I compute.

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